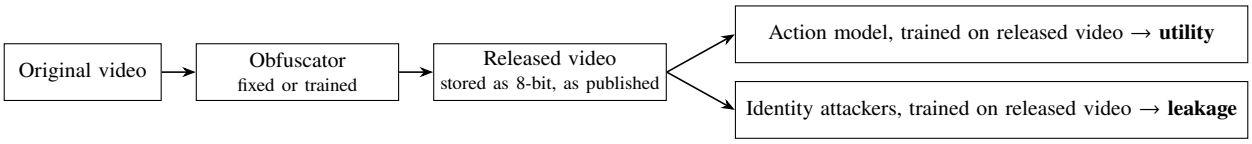
Concealing Identity in Action Video: System Architecture and Results

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Objective. To transform (*obfuscate*) video so that the action performed remains recognisable to a computer while the identity of the person does not. Each obfuscator is assessed by *utility*, the accuracy of an action classifier, and *identity leakage*, the accuracy of the strongest attacker at naming the person (top-1, in percent).



Evaluation principle. Every model is trained afresh on the obfuscated data, as anyone holding the released dataset could do. Prior work commonly uses attackers trained only on unaltered video; these fail because the input is unfamiliar, not because identity has been removed. Over 151 KTH runs they understated leakage by a median of 41 points: light blur appeared to reduce identification from 99.5% to 35%, whereas a retrained attacker still reached 99.5%.

1. Obfuscators examined

- **Fixed filters:** blur; pixelation; downscaling by k (at 16× a 120×160 frame becomes 7×10 pixels); frame differencing, which keeps only pixels that change between frames.
- **Learned filter (BDQ):** blur, frame difference, and rounding to 15 grey levels. The blur strength and the rounding thresholds (16 numbers) are learned. Downscaling may follow.
- **Learned redrawing:** a 3D convolutional network (0.6 M parameters) or an edit inside a frozen image autoencoder (3.6 M parameters) that repaints each frame.
- **Stick figures:** a pose detector finds 17 body joints, drawn as lines on a blank background. The *standardised* variant uses fixed limb lengths at a fixed size (removing height and build) and updates the drawing once per 16 frames (removing individual timing of movement), and is then downscaled. It requires no training.

2. Training of the learned filters

The filter is trained together with an action model and two adversarial identity models. It is rewarded when the action model succeeds and when the adversaries are uncertain, while the adversaries are trained to identify the person. The adversaries are reset every three epochs so the filter cannot specialise against one opponent. A safeguard keeps the last epoch whose action accuracy was within 5 points of its best, and stops training after three epochs more than 30 points below it.

3. Models and hyperparameters

item	setting	purpose
Models		
action model	R(2+1)D-18, Kinetics-400	video network that classifies the action
frame attacker	ResNet-50, ImageNet	identity from appearance; 4 frames per clip, 2× upscaled
video attacker	R3D-18, Kinetics-400	identity from motion and gait
check attacker	DINOv2 ViT-B/14	large general vision model, to confirm results
pose detector	Keypoint R-CNN, ResNet-50	17 joints; person confidence ≥ 0.7
Input		
clip	16 frames, every 2nd	≈1.2 s of KTH, 120×160 greyscale
test scoring	3 clips per sequence	predictions averaged
Training of evaluation models		
epochs, batch	12; 16 clips	passes over data; clips per update
optimiser	AdamW, lr 2×10^{-4}, w. decay 0.05	step size; decay limits overfitting
schedule	cosine, half-epoch warm-up	rate rises briefly, then decays
Adversarial training of the learned filter		
epochs, batch	15; 8 clips	
privacy weight α	3 with downscaling; 5–7 without	reward for confusing attackers vs. keeping the action
learning rates	filter 3×10^{-3}, action 10^{-4}, adversaries 2×10^{-4}	AdamW, weight decay 0.01
adversaries	ResNet-18 and R3D-18	reset every 3 epochs
blur, levels	$\sigma_0 = 2$ (range 0.3–8); 15 levels	starting blur; grey levels kept

4. Results on KTH

25 people, 6 actions (walk, jog, run, box, wave, clap), greyscale. Chance: 16.7% action, 4.0% identity. The *unseen scenario* columns test on a recording condition never used in training.

	standard split		unseen scenario	
obfuscator	action	identity	action	identity
original video	97.5	99.5	88.7	97.7
blur ($\sigma = 8$)	94.5	95.0	77.7	87.7
downscale 16×	91.8	92.6	–	–
frame difference	96.5	97.5	–	–
learned redrawing (autoencoder)	96.2	75.6	–	–
learned filter, no downscaling	89.0	50.0	60.3	31.4
learned filter → 3×5 px	91.9	40.0	62.4	28.7
learned filter → 1×2 px	80.9	11.4	52.0	13.3
stick figure	97.8	53.4	90.5	37.7
standardised stick figure	91.7	28.7	81.3	28.6
standardised stick figure, 16×	**89.1**	**13.7**	**73.3**	**18.7**

5. Principal findings

- Blur, pixelation and downscaling leave identity almost intact (92–99%); motion alone identifies people (97.5%).
- The standardised stick figure gives the best balance, 89.1% action against 13.7% identity, outperforming every learned filter. It holds under longer attacker training (88.3 / 13.9) and the DINOv2 attacker.
- Figures are means over 2–9 runs; learned filters vary by up to 6 points between runs.

6. Other datasets

dataset	best result	assessment
Charades (90 people, at home)	stick figure: identity 93.5→16.5; action mAP 15.8→6.6	**Partial.** The home identifies people; with people blacked out, identity is still 90.6.
UTD-MHAD (8 people, one lab)	motion map, 2 pixels: identity 100→55.4; action 100→62.9	**Unsuccessful.** Separate recording sessions: mean frame colour alone identifies 72.8.
VP-UCF101, VP-HMDB51	gender, skin colour etc.: marginal reductions	**Unsuccessful.** The attributes follow largely from the action label.

7. Prior work and limitations

Ilic et al. (CVPR 2024) reported KTH identity reduced to 5.5%. Their released code, re-run with 10 action and 5 identity models, reproduces the action accuracy (88%) but leaves identity at 78%; their attackers were never trained on obfuscated video. Our figures hold only against the attackers tested; KTH is a small laboratory dataset, and stick figures require visible bodies.